



# Data Analytics and Visualization (24DSB3101)

## An Explainable Machine Learning Framework For Predictive Financial Risk Assessment And Early Warning

*Under the Esteemed guidance of*

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# Introduction

- Financial institutions handle large amounts of customer, credit, loan, and transaction data. Traditional risk assessment methods may struggle to identify complex patterns within such data.
- Machine learning can analyze historical financial information and identify patterns associated with financial risk. In this project, different ML models will be compared to classify customers into **Low, Medium, and High-Risk** categories.
- The project also incorporates **Explainable AI (XAI)** to explain why a particular customer is classified as risky.



# Problem Statement

- Traditional financial risk assessment methods may fail to capture complex and changing patterns in financial data. Although machine learning improves prediction, many models behave like **black boxes**, making their decisions difficult for financial stakeholders to understand.
- The proposed system addresses this by providing:
- **Risk Prediction + Risk Score + Explainability + Early Warning**
- It aims to identify potentially high-risk cases and explain the major factors responsible for the prediction.



# Objectives

- Develop financial-risk prediction models using **Random Forest, XGBoost and CatBoost**
- Handle class imbalance using **SMOTE / Class Weights**.
- Perform **feature engineering and feature selection**.
- Apply **SHAP** to explain individual risk predictions.
- Generate a calibrated **financial risk score and risk category**.
- Provide an **early-warning mechanism** for potentially high-risk cases.
- Build an integrated **dashboard-based decision-support system**.

# Literature Survey



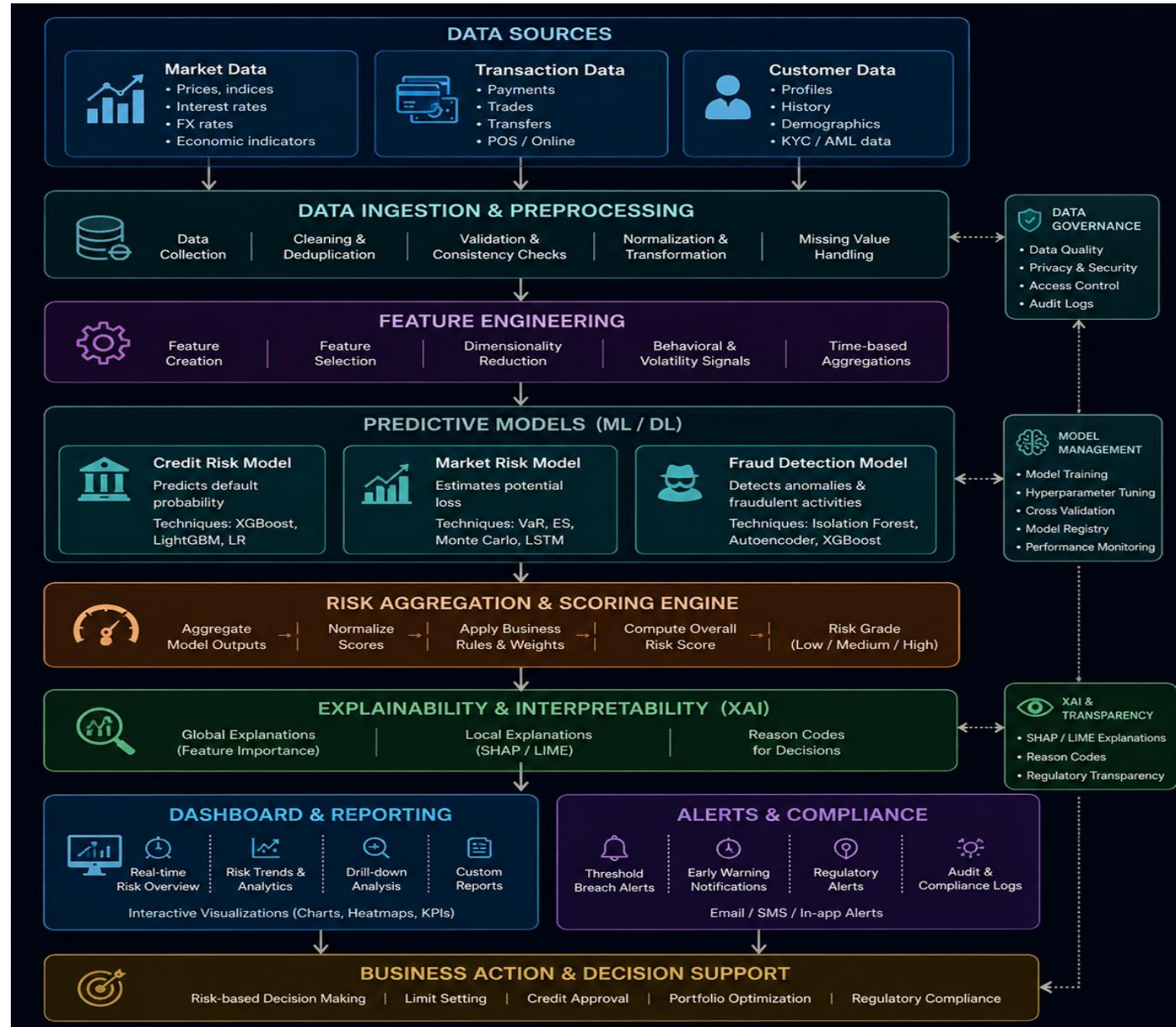
| No. | Year | Research Paper / Study                                                                              | Technique / Framework                                          | Key Contribution                                                            | Research Gap                                                                                 |
|-----|------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1   | 2026 | Enterprise Financial Risk Prediction Model Based on Machine Learning Algorithm                      | SVM, ANN, Logistic Regression / FinRiskNet                     | Uses multiple classifiers for enterprise financial-risk prediction          | Mainly prediction-focused; explainability and integrated early warning can be improved       |
| 2   | 2026 | Application of Generative AI in Corporate Financial Risk Early Warning                              | Multimodal Fusion, GNN, Temporal GAN, Interpretable Generation | Combines multiple data sources and generates risk reports/recommendations   | Complex architecture and high computational requirements                                     |
| 3   | 2025 | Enterprise Financial Risk Prediction Model Based on Modified Machine Learning Classifier Algorithms | Modified ML Classifiers                                        | Uses internal and global financial factors for prediction and early warning | Focuses more on accuracy than transparent decision support                                   |
| 4   | 2025 | Enterprise Financial Risk Prediction and Intelligent Early Warning Model Based on Deep Learning     | BiLSTM + Attention                                             | Captures temporal financial patterns and provides early warnings            | Deep-learning models are difficult to interpret and require sufficient temporal data         |
| 5   | 2025 | Enhancing Credit Risk Prediction through an Ensemble of Explainable Model                           | XGBoost + Multiple Explanation Methods + K-Means               | Combines ensemble prediction with explainability                            | Broader financial-risk scoring, risk trajectory and early-warning mechanisms can be explored |

# Literature Survey



| No. | Year | Research Paper / Study                                                                                          | Technique / Framework                        | Key Contribution                                                                 | Research Gap                                                              |
|-----|------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 6   | 2025 | Systemic Financial Risk Prediction Based on Machine Learning: The Role of FinTech                               | ML + FinTech Indicators                      | Uses a FinTech index to improve the timeliness of systemic-risk prediction       | Focuses on systemic risk rather than customer/enterprise-level risk       |
| 7   | 2025 | Graph-Based Inductive Learning for Credit Risk Prediction with Imbalance Mitigation                             | Graph-Based ML + Imbalance Mitigation        | Addresses class imbalance and relational information in credit-risk predict      | Requires complex relationship/graph data that may not always be available |
| 8   | 2025 | Sustainable Credit Risk Prediction: Applying Machine Learning for Responsible Lending                           | Feature Engineering + Ensemble ML + CatBoost | Applies ensemble ML to credit-risk prediction and responsible lending            | Explainable early-warning integration remains an opportunity              |
| 9   | 2024 | <b>Prediction of Bank Credit Worthiness Through Credit Risk Analysis: An Explainable Machine Learning Study</b> | Explainable Machine Learning                 | Uses ML and XAI to predict bank creditworthiness                                 | Can be extended toward broader risk scoring and early-warning mechanisms  |
| 10  | 2024 | Financial Risk Forewarning with an Interpretable Ensemble Learning Approach                                     | Bagging + LightGBM + SHAP + MOPSO            | Combines ensemble learning, optimization and SHAP for financial-distress warning | Primarily focused on listed-company financial distress                    |

# Project Workflow





# Research Gap

## 1. Fragmented Risk Management

Existing research often concentrates on individual risk types rather than a flexible framework.

## 2. Prediction vs. Explainability

High-performing models such as XGBoost can be difficult to interpret.

## 3. Limited Risk Scoring

Many systems provide only binary predictions such as Default/Non-Default.

## 4. Limited Early Warning

Prediction and continuous risk monitoring are often treated separately.

## 5. Class Imbalance

High-risk/default cases are usually much fewer than low-risk cases.

## 6. Limited Decision Support

Many systems do not combine:

## 7. Risk Score → Risk Level → Important Factors → Recommended Action

These six gaps are specifically identified in your review document.



# Novelty

- **Explainable Risk Prediction:** Integrates Explainable AI techniques such as **SHAP/LIME** with machine learning models to clearly identify the key factors contributing to an individual's or organization's financial risk.
- **Early Warning System:** Develops a **predictive early-warning mechanism** that detects increasing financial risk before major financial problems occur, enabling timely preventive decisions.
- **Hybrid & Integrated Framework:** Combines **multiple ML models, financial indicators, and explainability techniques** into a unified framework to improve prediction accuracy, transparency, and decision-making compared with traditional financial risk assessment methods.

# Innovation



- Machine Learning Prediction



- Risk Score



- Risk Classification



- SHAP Explanation



- Early Warning



- Recommended Action

# Feasibility



- **Technical Feasibility:** The system can be developed using Python, Scikit-learn, XGBoost/LightGBM, SHAP, Imbalanced-learn, and Streamlit.
- **Dataset Feasibility:** Public financial datasets such as German Credit, Give Me Some Credit, and Lending Club Loan Dataset are available for experimentation.
- **Computational Feasibility:** Ensemble ML models can be trained on normal academic computing systems without requiring expensive deep-learning infrastructure.
- **Implementation Feasibility:** Open-source tools can support data preprocessing, risk prediction, explainability, early warning, and dashboard development.

# Project Plan



**Phase 1:** Literature Survey & Research Gap Identification



**Phase 2:** Dataset Collection & Preprocessing



**Phase 3:** Feature Engineering & Feature Selection



**Phase 4:** Class Imbalance Handling using SMOTE / Class Weights



**Phase 5:** Baseline Model Training



**Phase 6:** Ensemble Model Development using XGBoost / LightGBM



**Phase 7:** Model Evaluation & Comparative Analysis



**Phase 8:** SHAP-Based Explainable AI



**Phase 9:** Risk Score & Early-Warning Development



**Phase 10:** Interactive Dashboard Development



**Phase 11:** Testing & Performance Validation



**Phase 12:** Documentation & Final Presentation



THANK YOU

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